

LECTURE ACTIVITY 7: DO PEOPLE USE FACIAL PROTOTYPING? – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from “Introduction to Statistical Investigations,” by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

- 1) Calculate and interpret the standardized statistic under a simulation-based approach
- 2) Calculate and interpret the standardized statistic under a theory-based approach
- 3) Conduct a theory-based approach when evaluating claims about a single population proportion.
- 4) Identify whether the validity conditions for a one-sample z-test are met.
- 5) Evaluate the strength of evidence against the null hypothesis based on the p-value and/or standardized statistic (z)
- 6) State a conclusion about the alternative hypothesis (and null hypothesis) based on the magnitude of the standardized statistic and/or p-value.

Key Terms:

- | | | |
|---------------------|--------------------------|-------------------------|
| • Null distribution | • Standardized statistic | • Normal approximation |
| • p -value | • Validity conditions | • Central Limit Theorem |
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Part 1: Review of Simulation-Based Inference

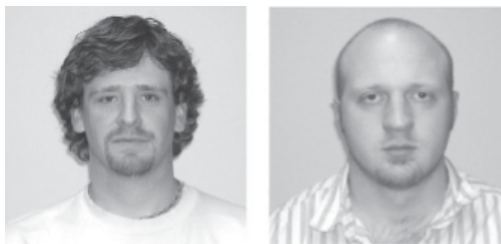


Figure 1: Melissa A. Lea/Robin D. Thomas/Nathan A. Lamkin/Aaron Bell ©Psychonomic Bulletin & Review 2007, 14 (5), 901–907.

A study in Psychonomic Bulletin and Review (Lea, Thomas, Lamkin, and Bell, 2007) presented evidence that “people use facial prototypes when they encounter different names.” Participants were given two faces and asked to identify which one was Tim and which one was Bob. Figure (1 to the left depicts the faces that were shown to the subjects in the study. The researchers wrote that their participants “overwhelmingly agreed” on which face belonged to Tim and which face belonged to Bob but did not provide the exact results of their study.

Step 1: State the Research Question

Shown above are the two photos in question. The question we’re interested in is: are students in my three lecture sections more likely to identify Tim as the picture on the left?

Step 2: Design a Study and Collect Data

The “data collection survey” I sent out to my three STAT 1110 lecture sections asked you to pick which face you think went with which name. The **parameter** of interest in this study is the population proportion of

students who would assign the name Tim to the picture on the left.



Question 1

(a) Describe an observational unit in the context of this study.

Solution: The way the parameter is phrased (as a population proportion), we should think of one observational unit as **one person** (more specifically, **one student**).

(b) Describe the variable in the study. Is it quantitative or categorical? If it is categorical, what are its levels?

Solution: The variable is **which name the student assigned to the picture on the left**. It is a **categorical** variable, with levels **Tim on Left** or **Bob on Left**. Notice that we've encoded this as a **binary** variable.

When we conduct analyses with binary variables, we often call one of the outcomes a “success” and the other a “failure” and then focus the analysis on the “success” outcome. Sometimes, we try and match the notion of a success to a *natural* one (like Harley correctly picking a cup), but we don't always have to match the “natural” notion of success. Just make sure that you are consistent with your notion of a success - don't switch partway through your analysis!



Question 2

State the null and alternative hypotheses each in two different ways: once using **words**, and then again using **symbols**.

Solution:

- **Null Hypothesis:** the population proportion of students that would assign Tim to the name on the left is 0.5 (“just guessing”).
 - Letting π_{Left} denote the population proportion of students that would assign “Tim” to the name on the left, then we can express our null hypothesis as $H_0 : \pi_{\text{Left}} = 0.5$.
- **Alternative Hypothesis:** the population proportion of students that would assign Tim to the name on the left is greater than 0.5 (“more than by-chance-alone”).
 - Letting π_{Left} denote the population proportion of students that would assign “Tim” to the name on the left, then we can express our null hypothesis as $H_0 : \pi_{\text{Left}} > 0.5$.

A couple of notes:

- Remember to always verbally define your parameters before using them. For example, just writing $H_0 : \pi_{\text{Left}} = 0.5$ would not be correct without first defining what π_{Left} represents.
- Also remember that hypotheses are conjectures about parameters: it is incorrect to write $H_0 = 0.5$, because a hypothesis is never “equal” to a value.

Step 3: Explore the Data

Shown below is a summary of the responses I got from the “data collection survey.” A total of 68 students responded, of which 52 assigned the name “Tim” to the picture on the left. See the sample distribution of the class data below:

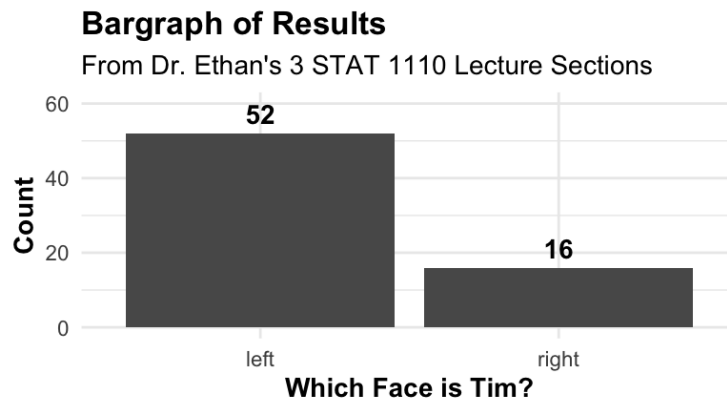


Figure 2: Results from the data collection survey. Of the 68 respondents, 52 assigned the name “Tim” to the photo on the left and the remaining 16 assigned the name “Tim” to the photo on the right.



Question 3

- (a) The sample size is equal to 68
- (b) The value of the statistic (sample proportion of students who put Tim’s name on the left) is equal to $52 / 68 \approx 0.7647$
- (c) The symbol we use for the quantity in part (b) above is (select one of the following options):
- ☐ π ☒ $\hat{p}$

Step 4: Draw Inferences

So we can get some more practice with the **simulation-based** side of things, let’s use the **One Proportion Applet** (for those of you with a printed copy of this handout, here is the url: <https://www.rossmanchance.com/applets/2021/oneprop/OneProp.htm>) to carry out our inference. In the second part of today’s lecture activity, we’ll explore a different approach!



Question 4

Fill in the blanks below with the appropriate values/options to select in the applet to ensure we get out 10,000 repetitions of the simulation.

- (a) Probability of success = 0.5 (**we want this to match the null value; i.e. the value of the parameter under the null**)
- (b) Number of tosses = 68 (**we want this to match sample size**)
- (c) Number of repetitions = 10,000 (**we’re told we want to get out 10,000 repetitions of the simulation**)

Now, **conduct the simulation!** Make sure the “Proportion of heads” button is selected in the applet (not the “Number of heads” button). This ensures we’re on a scale consistent with *proportions*, and not *counts*.

Do not close out of the applet yet - you'll need to make some modifications to your plot in the next few questions!

Question 5

In this question, we'll work toward using the applet to calculate an approximate p -value.

- (a) Under the “Count samples” option, which sign should we use? (Select one.)

☐ $\leq$ ☐ $=$ ☒ $\geq$

- (b) Under the “Count samples” option, the number we should put in the empty box is 0.7647
(this should be the value of $\hat{p}$ we observed in our actual data)

- (c) After selecting the appropriate option and filling in the appropriate blank, click “count” and report your approximate p -value below.

- My approximate p -value is 0.00 (note that your answers may be a little different than mine, due to sampling variability)

Insert a screenshot of your entire screen, showing the options you put in, below:

Solution: Answers may vary. Here's my plot:

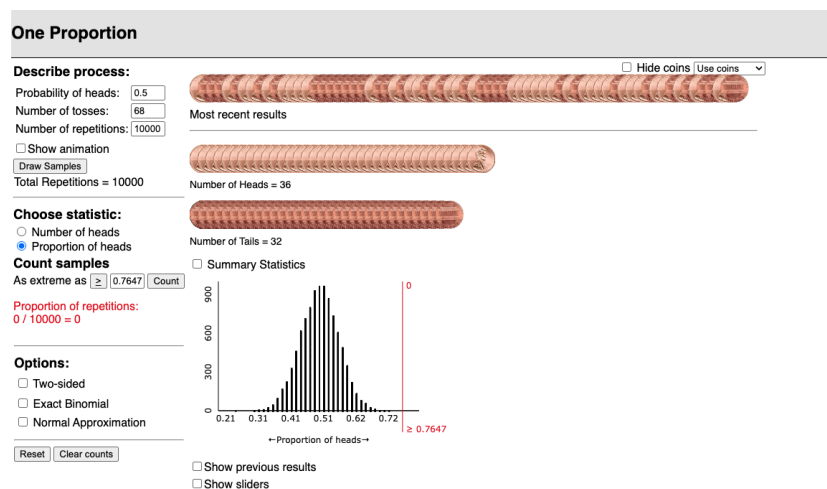


Figure 3: Screenshot of the applet, with the correct inputs.

The resulting distribution (of simulated $\hat{p}$ values under the null) is a simulation of the **null distribution**.

Question 6

What does one dot on the dotplot above (your simulated null distribution) represent? Be as specific as possible.

Solution: One dot represents a simulated proportion of students (in a hypothetical sample of 68 STAT 1110 students) who assigned the name “Tim” to the photo on the left.

Alright, let's go back to our p -value. Remember (from the last lecture activity) that there are two things we

need to do with a p -value (approximated or otherwise) - **evaluate** and **interpret**.

 Question 7

- (a) **Evaluate** this approximate p -value. Based on the guidelines established in the previous lecture activity (which matches the guidelines from the textbook), how much evidence does the sample data provide against the null hypothesis? Choose one.
- A. Not much evidence against the null hypothesis
 - B. Moderate evidence against the null hypothesis
 - C. Strong evidence against the null hypothesis
 - D. Very strong evidence against the null hypothesis
- (b) **Interpret** this approximate p -value. Please note that interpretation is not the same as evaluation! I have added a “skeleton” interpretation to help you:

The approximate p -value of 0.00 is the approximate probability of observing a sample proportion (specifically, a sample proportion of students who put Tim’s name on the left) of 0.7647 or greater, under the assumption that people are equally likely to put Tim on the left or right.

Some additional guidance/hints:

- The first blank should be your answer to Question 5(c)
- The second blank should be your answer to Question 3(b)
- For the third blank, choose either “greater” or “smaller” (only one of these is correct!)
- For the third blank, choose either “equally likely” or “less likely” (only one of these is correct!)

The p -value is the most common way to evaluate strength of evidence against the null hypothesis, but now we will explore a common alternative way to evaluate strength of evidence. The goal of any measure of strength of evidence is to use a number to assess whether the observed statistic falls in the tail of the null distribution (and is therefore surprising when the null hypothesis is true) or among the typical values we see when the null hypothesis is true.

 Question 8

Navigate back to the applet, and check the “Summary Stats” box. What this does is calculate summary statistics of the 10,000 simulated $\hat{p}$ values.

- (a) Report the mean value of the simulated null distribution: 0.500.
- (b) Explain why it makes sense that the reported mean is close to 0.50. **Hint:** think about what assumption “under the null” really means. In other words - what assumption did we make when conducting our simulation?

Solution: “Under the null” means “assuming the null distribution is true.” That is, we conducted our simulation by explicitly assuming the true proportion of people who would assign “Tim” to the left is 0.5! As such, it makes sense that the average (mean) of the simulated $\hat{p}$ values would be around 0.50.

- (c) Report and interpret the standard deviation of the null distribution (SD_{null}). 0.061

Solution: Answers will likely vary from simulation to simulation. But, the standard deviation of the simulated null distribution should be around 0.06.

- (d) Calculate how many standard deviations the observed class value of the statistic is from the hypothesized mean of the null distribution (0.50). In other words, subtract 0.50 from the observed statistic (proportion of students who put Tim’s name on the left) and then divide by the standard deviation of the null distribution (SD_{null}). In other other words, calculate

$$\frac{\hat{p} - 0.50}{SD_{\text{null}}}$$

Solution: $\frac{\hat{p} - 0.50}{SD_{\text{null}}} = \frac{0.7647 - 0.50}{0.061} \approx 4.34$.

The observed statistic $\hat{p}_{\text{Left}} \approx 4.34$ is about **4.34** standard deviations **above** the null value of 0.5.

Concept Overview

The value we just calculated is called the **standardized statistic**, and we use the symbol z to represent it:

$$z = \frac{(\text{statistic}) - (\text{hypothesized mean of null distribution})}{(\text{Standard Deviation of Null Distribution})}$$

Once you calculate this value, we can interpret it as *how many standard deviations the observed statistic falls above/below the hypothesized parameter value*. We can use the standardized statistic to evaluate the strength of evidence against the null hypothesis (just like we did with the p -value).

Pro Tip

Another way to think about using the standardized statistic is as a slightly more rigorous way of quantifying whether the observed statistic is in the tail or near the center of the null distribution (like we started out doing, in Lecture Activity 5).

The figure below shows the positions of standardized statistics for a bell-shaped distribution. The figure can be summarized by the following key idea: observations that fall more than 2 or 3 standard deviations from the mean can be considered “in the tail” of the distribution (unlikely under that distribution).

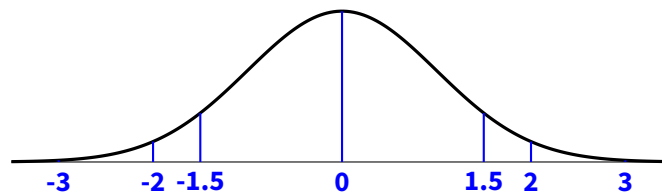


Figure 4: An illustration of how values that fall more than 2 or 3 SDs from the mean can be considered “in the tail” of the distribution

Based on the plot above, we can use the following guidelines as the basis for using a standardized statistic to assess the strength of evidence against the null hypothesis for a mound-shaped, symmetric distribution.

Absolute value of z	Strength of Evidence
Between 0 and 1.5	Not much evidence against the null hypothesis; null is plausible
Between 1.5 and 2	Moderate evidence against the null hypothesis
Between 2 and 3	Strong evidence against the null hypothesis
Above 3	Very strong evidence against the null hypothesis

Table 1: Guidelines for evaluating the strength of evidence from a standardized statistic (z)

⚠ Caution

Just like with the table about p -values, these are only **suggested guidelines**.

💡 Pro Tip

Your textbook has a slightly different table, in that it does not differentiate between negative and positive standardized statistics. In my opinion, though, the sign of the standardized statistic is still important - it tells us which side of the hypothesized mean of the null distribution our statistic will be! But, in terms of strength of evidence, the larger the *absolute value* of the statistic, the stronger the evidence against the null and in favor of the alternative.

✍ Question 9

- (a) Based on the value of the standardized statistic (z) in Question 8, how much evidence do the class data provide against the null hypothesis? Choose one.

- ☐ Not much evidence against the null hypothesis
☐ Moderate evidence against the null hypothesis
☐ Strong evidence against the null hypothesis
☒ **Very strong evidence against the null hypothesis**

- (b) How closely does your evaluation of strength of evidence based on the standardized statistic compare to the strength of evidence based on the approximate p -value in Question 5(c) above?

Solution: The evaluations match closely. Both the very small p -value and the large z -statistic show very strong evidence against the null hypothesis.

Step 5: Formulate Conclusions

The research question we explored in this activity was whether students are more likely to identify Tim as the picture on the left.

Question 10

Answer the research question. Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) p -value, (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p -value, (4) reject/fail to reject the null hypothesis, and (5) overall conclusion in context. Make sure you refer to the **parameter** in your answer.

Solution: Based on the observed statistic of $\hat{p}_{\text{Left}} = 52/68 \approx 0.7647$ representing the sample proportion of students who chose Tim as the person on the left, we obtained a p -value **0.00**. This constitutes very strong evidence against the null hypothesis leading us to reject it. We have very strong evidence that the population proportion of students who would put Tim on the left is greater than 0.50. In other words, students like the ones to whom I administered the survey are more likely to put Tim on the left.

As a reminder, the sample in this study was composed of the students in your class: STAT 1110 students with ages between 18 – 20. Even though this was not a random sample, we can still generalize the results of this study, but we need to do it with caution.

Question 11

- (a) Do you think we can apply the results of this study to students outside the US?

Solution: We should be cautious applying the results to students outside the US because there might be a cultural reason for why students put Tim on the left and this might not be true in other countries.

- (b) Do you think we can apply the results of this study to people in San Luis Obispo in their 40s?

Solution: We should also be cautious applying the results to people in San Luis Obispo in their 40s because the sample consisted of college students ages 18–20. We do not know if people in their 40s would behave in the same way regarding the pictures as students between 18 and 20 years old.

Step 6: Look Back and Ahead

Question 12

Based on the limitations of this study, suggest a new research question that you would investigate next.

Solution: A good next research question might be to look at different pictures and different names.

Part 2: Theory-Based Approach (One-Proportion Z-Test)

The framework for hypothesis testing we've used up until now (in the past few lecture activities) is often classified as **simulation-based**. It gets this name based on our reliance on *simulations* to artificially generate “new” values of $\hat{p}$ under the null. The simulation-based approach is highly flexible and quite interpretable, but also suffers from some drawbacks. In this section, we'll try and make use of more general results to help us carry out our inference in a slightly different fashion.

You might have noticed that a lot of the simulated null distributions we've encountered in this course thus far follow a similar pattern: they're unimodal, symmetric, and bell-shaped. Furthermore, they've all been (roughly) centered at the null value (i.e. the value of the parameter, assuming the null is true). This is not an accident! There is a theoretical result (that we'll discuss in a moment) that justifies this pattern.

We use mathematical models known as **normal distributions** (bell-shaped curves) to approximate many of the null distributions we have generated so far in this text. When rules and theories are used to predict what the value of the standardized statistic and p -value would be if someone did simulate, we call the approach a **theory-based approach**. The normal distribution provides a second way, in addition to simulation, to approximate a p -value.



Question 13

Navigate back to the One Proportion applet, and check the box next to “Normal Approximation.” This will overlay a normal distribution curve on top of the dotplot. Does the region shaded in blue (the normal approximation) seem to be a good description (model) of what we actually got in the simulation (the dotplot)?

Solution:

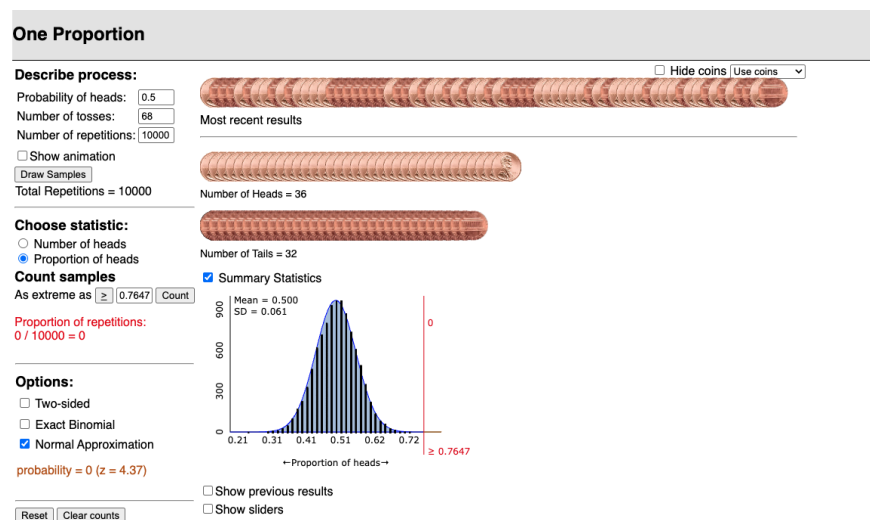


Figure 5: A screenshot from the applet, with the approximating normal distribution curve overlaid.

Concept Overview

The normal approximation can be thought of as a prediction of what would occur if a simulation-based analysis was carried out. Many times this prediction is valid, but not always. The normal approximation

to the null distribution is valid whenever the sample size is reasonably large. One convention is to consider the sample size to be large enough whenever there are **at least 10 observations in each category (successes and failures)**.

 Question 14

According to this convention, is the sample size large enough in this study to use the normal approximation of the simulated null distribution and theory-based inference? Justify your answer by referring to the sampled data.

Solution: Yes; we have at least 10 successes (52 students assigning “Tim” to the left) and at least 10 failures (16 students assigning “Tim” to the right).

Statistic

In this part of the activity, we will use the standardized statistic z to calculate the p -value under a theory-based approach.

Concept Overview: Formulas under the Theory-Based Approach

The normal approximation will also give you values of the standardized statistic and p -value based on its mathematical predictions. The standardized statistic is calculated as

$$z = \frac{(\text{statistic}) - (\text{hypothesized mean of null distribution})}{(\text{Standard Deviation of Null Distribution})}$$

The mean of the null distribution is the null value π_0 (i.e. the hypothesized value of the population proportion π , under the null). The standard deviation can be obtained in two ways:

- **Option 1:** find the standard deviation of the null distribution by simulating
- **Option 2:** predict the value of the standard deviation by plugging into this formula:

$$\sqrt{\frac{\pi_0(1 - \pi_0)}{n}}$$

- Note: when conducting the z -statistic we are under the null hypothesis, so the value of π we should use is whatever we stated under the null hypothesis (which I am representing as π_0).

 Question 15

- (a) Use the formula $\sqrt{\pi_0(1 - \pi_0)/n}$ to calculate the theoretical/predicted standard deviation of the sample proportions.

Solution: $\sqrt{\pi_0(1 - \pi_0)/n} = \sqrt{0.5 \cdot (1 - 0.5)/68} \approx 0.0606$.

- (b) Compare this value with the one you obtained from the simulated sample proportions, in Question 8(c). Are they similar?

Solution: Yes, they are similar! Again, this is expected since the validity conditions are met - we have at least 10 successes (namely, 52) and at least 10 failures (namely, 16) so the theory-based approach tells us that the predicted standard deviation of the null distribution should be around 0.0606. Hence, we would expect the sample standard deviation of a large number of simulated $\hat{p}$ values, under the null, to be around 0.0606.

The predicted value of the standard deviation (using the formula) will be very close to the simulated standard deviation of the null distribution. The validity condition mentioned earlier says the shape of the null distribution will be approximately normal when the sample size is large enough where, a “large enough” sample size means at least 10 successes and at least 10 failures. This mathematical prediction is often called the “**central limit theorem**.”

Concept Overview: The Central Limit Theorem

If the sample size (n) is large enough, the distribution of the sample proportions will be bell-shaped (or normal), centered at the population proportion (π), with a standard deviation of $\sqrt{\pi(1 - \pi)/n}$.

Question 16

Use the theoretical value of the standard deviation you just computed to calculate the standardized statistic (z) by hand and confirm that your answer is very close to the z statistic you found when using simulation. Show your work step by step.

Solution:
$$z = \frac{\hat{p} - \pi_0}{\sqrt{\pi_0(1 - \pi_0)/n}} = \frac{0.7647 - 0.5}{0.0606} \approx 4.368$$

Strength of Evidence

In the applet, see that the predicted value of the standardized statistic, z , is given immediately below the button for “Normal approximation” in parentheses and should match your answer to the previous question.

Question 17

The theory-based (normal approximation) p -value is also now displayed. Compare this p -value to the one you got from simulation. Are they similar? To show your work, include a **screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) p -value**. Tip: make sure that the Normal curve in blue is showing in your screenshot.

Solution:

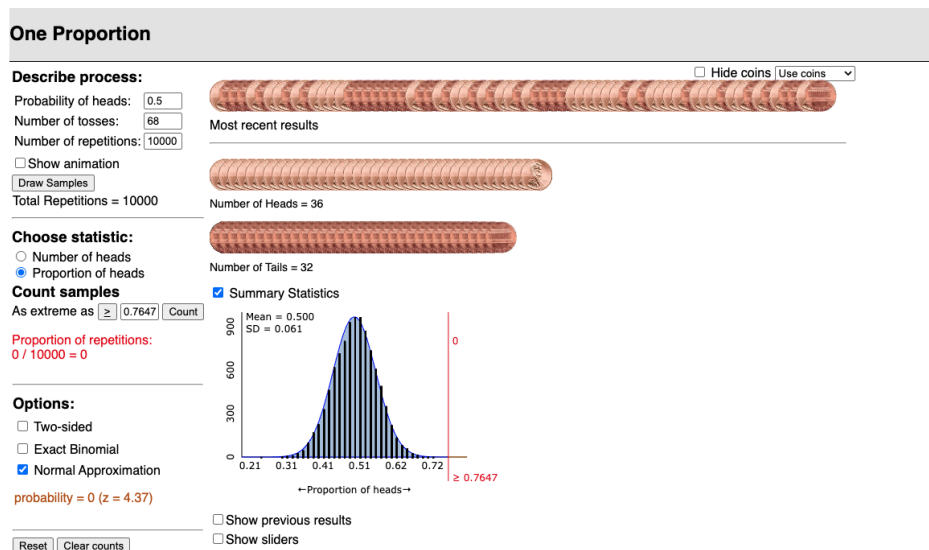


Figure 6: A screenshot with all relevant inputs and outputs, and the approximating normal distribution curve.

Pro Tip

For now on, we will **not** use the One Proportion applet to conduct the theory-based approach. Instead, we will use the **Theory-based Inference applet**.

Question 18

Carry out the following steps:

- 1) Navigate to the **Theory-Based Inference applet** (for those of you with printed copies of this handout: <https://www.rossmanchance.com/applets/2021/tbia/TBIA.html>).
- 2) Use the pull-down menu to select “One Proportion.”
- 3) Enter the sample size and count
- 4) Click “Calculate”
- 5) Check the “Test of Significance” box
- 6) Input the correct value and sign for the hypothesis
- 7) Press “Calculate”

Include a screenshot of your entire screen below, showing any/all inputs and outputs you specified.

Solution:

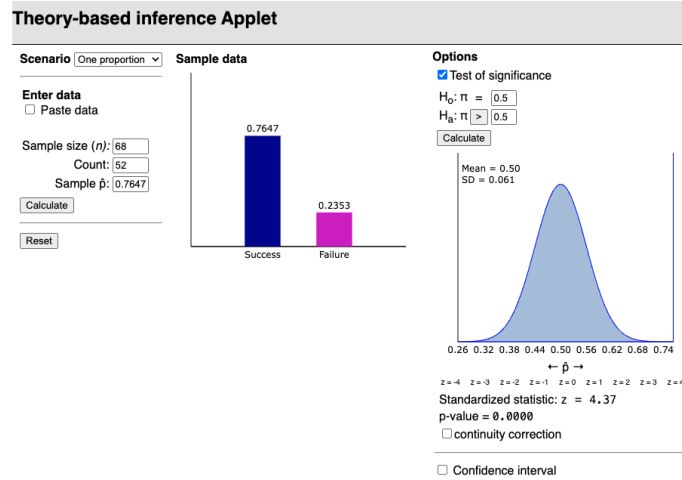


Figure 7: A screenshot from the Theory-Based Inference Applet, with correct inputs.